Empire AI: A new model for provisioning AI and HPC for academic research in the public good

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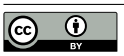


Figure 1: The Empire AI Alpha system, October 2024.

Abstract

Launched in April 2024 by New York State Governor Kathy Hochul, Empire AI is a consortium of New York's leading public and private universities collaborating to establish a state-of-the-art artificial intelligence computing center, to be housed at SUNY's University at Buffalo. We describe this new national model in ethical AI innovation and how it is pioneering safe, equitable, and accessible AI research and development that is benefiting every corner of New York. Our technology and facility roadmap is presented along with preliminary estimates of economic impacts. In summer 2025, Empire AI will be the first academic deployment of an NVIDIA DGX GB200 NVL72 SuperPOD that will have 288 B200 GPUs delivering exascale performance in 16-bit tensor operations and able to train multi-trillion parameter models.

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CCS Concepts

• **Computer systems organization**; • **Social and professional topics**; • **Applied computing**;

Keywords

artificial intelligence, high-performance computing, supercomputing

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1 Introduction

Empire AI Consortium, Inc., is a not-for-profit company incorporated in New York State in July 2024 *"to create and launch a state-of-the-art artificial intelligence computing center in Upstate New York to be used by New York's leading institutions to promote responsible research and development, create jobs, and unlock AI opportunities focused on public good [9]."* Specifically, Empire AI seeks to address the dominance by large technology companies of the AI ecosystem due, in part, to the massive scale of computer resources available to them. This is in sharp contrast to the past when, for decades, it was academia at the forefront of computation due to supercomputers operated by nations for open scientific research. As a result, academic research in both the foundations and applications of AI is lagging, with implications for AI safety and society at large. By addressing this disparity, we enable trusted, in-state academic institutions to direct AI capabilities towards public benefit rather than relying upon the good intentions of external commercial entities. We also position our institutions to be nationally and internationally competitive whether recruiting faculty and students or capturing research funding. In turn, *"this pioneering AI research and development will also help educational institutions incubate the AI-focused technology startups of the future, driving job growth [9]."*

The Consortium presently includes City University of New York (CUNY), Columbia University, Cornell University, New York University (NYU), Rensselaer Polytechnic Institute (RPI), and State University of New York (SUNY), with the Simons Foundation as a close partner. In April 2024, the state committed $250M in capital funds that is complemented by $150M of contributions through these academic institutions for operations over the first decade. Pending approval in the State budget is an additional $90M to bring in 3 new members and provide an additional allocation to SUNY. Thus, we anticipate the final budget being well over $0.5B Initially scoped as a 10-year project, the consortium already has a much longer-term vision. Empire AI's board of directors includes a member from each institution, and 3 independent and 2 advisory members. The chair is Tom Secunda [2], whose vision and philanthropy have been essential to the establishment of Empire AI. The initial capital investments will be the purchase of an interim computer, and construction of a new data center. These and future plans are described in more detail below, including our initial computer resource (Alpha) and its upcoming major upgrade (Beta).

2 Alpha system

Alpha is on loan to Empire AI as a philanthropic gift from the Simons' Foundation [8] and is generously operated by staff of their Flatiron Institute [7]. It is housed at the University of Buffalo (UB) Center for Computational Research (UB-CCR). The air-cooled Lenovo HGX system comprises: (1) 13 HGX nodes each with 8 H100 80GB GPUs, 2 TB memory, and 30TB NVMe; (2) Dual 400Gb/s non-blocking NDR Infiband networks; (3) 2 login nodes; (4) 2 cluster management nodes for NVIDIA Base Command; (5) 2 data transfer nodes; (6) 2.8PB of DDN flash storage for training data, snap shots; and (7) 2PB effective of VAST flash storage for home directories.

3 Current research activities using Alpha

Over 300 proposals were submitted to use Alpha but, due to the limited resource, only 90 projects were started. Starting April 2025, a new cohort of projects is to be onboarded. The following categorization of active projects is edited from content generated by Gemini Advanced 2.0 Flash from the project descriptions (the only use of generative AI in this paper):

- **Foundation Models and Generative AI:** Developing and improving large-scale AI models that can generate various types of data (text, images, code, etc.) and serve as a base for more specialized applications.
- **AI for Scientific Discovery and Simulation:** Leveraging AI and machine learning to accelerate scientific research across various disciplines, including developing surrogate models for complex physical simulations.
- **AI in Healthcare and Biomedical Research:** Applying AI to improve healthcare, diagnostics, and our understanding of biological systems, including models for medical image analysis, biomolecular structure prediction and drug design, or creating personalized AI for assistive communication and mental health.
- **Natural Language Processing and Understanding:** Advancing the capabilities of AI to process, understand, and generate human language including analysis of hate speech, and exploring the ethical implications of LLMs.
- **Robotics and Embodied AI:** Developing intelligent agents that can interact with the physical world including creating robots with improved perception and AI-powered interactive speech therapy agents.
- **Machine Learning Methodology and Applications:** Research on fundamental machine learning techniques and their application to diverse problems not categorized above.

We highlight one project from each institution:

- **Columbia University:** Prof. Tian Zheng is building models of the climate, which can help communities prepare for future impact of extreme weather events.
- **Cornell University:** Prof. Ekta Khurana at Weill Cornell Medicine will use Empire AI to improve prostate cancer diagnosis and treatment, helping patients in New York State and reducing healthcare costs.
- **CUNY:** Prof. Rabinda Mandal is studying how malaria affects the gut microbiome, pioneering new strategies to combat infectious diseases.
- **NYU:** Prof. Christine Constantinople is using AI to better understand the human brain, in particular how we make decisions and solve problems.

RPI: Profs. Ge Wang and Pingkun Yan are building models to analyze CT scans to better diagnose/treat lung cancer.

SUNY: Prof. Srirangaraj Setlur and team at UB are developing technologies that will provide adaptive speech and language therapies to children with special needs.

3.1 Proposals, export control, allocations, and scheduling

Only faculty, staff, postdocs, and students of member institutions who live in the US are allowed to use Empire AI resources, and consortium members approve all projects and teams. Direct access to Empire AI systems is blocked from countries of concern including China. Project proposals are centrally collected by Empire AI using a uniform format and process, and then member institutions perform their own screening and prioritization. These processes and policies follow academic norms including those of NSF ACCESS [1, 3], and are strongly aligned with federal requirements including NIST frameworks, NSPM-33 [14], export control laws and trade compliance, and OSTP guidelines. Note that Empire AI is presently conceived to only support unclassified, fundamental research as defined in NSDD-189 [6] that is exempt from export control restrictions; indeed, several of our member institutions *only* perform such research. Nevertheless, Empire AI and its consortium members already have strong and comprehensive protections that help prevent export control violations. As detailed by the AAU [13], these include: compliance trainings and security requirements, monitoring and disclosure requirements on university staff and faculty, and coordination with Federal law enforcement to investigate issues.

For the upcoming Beta system (see below), we anticipate soliciting several classes of proposals, including "capability" (e.g., to train massive models) and "capacity" (e.g., high-throughput workflows). On Alpha we experimented with scheduling policies to achieve efficient and equitable system use while also facilitating large jobs to use all available resources. Each institution has its own Slurm partition. Initially, each week a different institution would have high priority on 12 of the 13 nodes in Alpha. As intended, this achieved equity between institutions and also reduced the drain time associated with scheduling large jobs. However, often only a few users dominated usage in a priority week and relatively few large jobs were run since workloads were still ramping up. Also, with six (and potentially soon nine) member institutions, users would have a long wait for their priority week, which was hard to align with the natural schedule of research. So we now allocate just six of the nodes in a priority week, with the remaining nodes open to all, and with fair-share scheduling within all Slurm partitions. We do not anticipate needing the priority week on Beta since it will be much larger and we aim to achieve equity between institutions through finite project allocations, fair-share and, hopefully minimal, manual priority adjustments.

4 Establishing a community and crowd-sourcing operations

Alpha is generously operated by the Flatiron Foundation team led by Dr. Ian Fisk. Already serving about 250 users and growing, the work load is substantial. Even when we have a full Empire AI staff, there will be a challenge meeting the full scale and diversity of users and workflows, including for user training. Following the campus champion model [4], we identified at each institution a technical point-of-contact (PoC) who is typically a senior person in the local research-computing or IT department. Each site also identifies several agents (typically scientific research support staff) who can login in to Empire AI computers and about two of whom will access our ticket system. These agents are charged with primarily supporting users from their own site, whom they presumably already know and support, as well as building up documentation, etc., used by all sites. All PoCs and agents meet virtually every two weeks, to discuss issues and implementation choices, etc. The PoCs and agents are now being trained in the containerized software technologies to be deployed on Beta, and will then scale that training to the full user base.

5 Beta system

Beta serves three main roles: (1) substantial compute power including both capability and capacity; (2) technology exploration for Gamma; and (3) establishing Empire AI Consortium's leadership by being the first academic initiative to employ this advanced hardware and software technology. Beta will start operations in early fall 2025 and will be temporarily located with Alpha in the UB-CCR. Its NVIDIA Blackwell GPUs are faster with more memory than those in Alpha. Once software and user workflows are fully tuned, Beta should be up to ~7x faster than Alpha on ML training and up to ~20x on inference. The DGX GB200 NVL72 SuperPOD technology tightly integrates the GPUs, enabling efficient training of multi-trillion parameter models. Each rack of the liquid-cooled NVL72 consumes up to 135kW power with up to 20% released to air. The Grace-Grace nodes (~ 500kW each) will also be liquid cooled. UB is making substantial power upgrades to accommodate Beta and also temporarily relocating equipment. Both Alpha and Beta will be relocated to the new facility (see below) so that UB can recoup their investment. Beta will include:

- An NVIDIA DGX GB200 NVL72 SuperPOD with 288 B200 GPUs for AI applications;
- 60 nodes of NVIDIA Grace-Grace Superchip for HPC and data processing;
- 20+ Petabytes effective flash storage from VAST as our primary advanced data platform;
- 10+ Petabytes flash storage from DDN for high-performance data
- The NVIDIA Run:ai and AI Enterprise commercial-grade AI software for training and inference; and
- 100 Gbit/s connection to NYSERNET [12], the New York State Education and Research Network.

The SuperPOD will include ~70 TByte of LPDDR5X memory at up to 18.4 Tbyte/s and ~54 TByte of HBM3e memory at up to 512 Tbyte/s. The tensor cores will achieve 5.76 exaFLOP/s in FP4, 2.88 in FP8, 1.44 in BF16/FP16, and 0.72 in FP32. Non-tensor performance is 23.0 petaFLOP/s in FP32 and 11.0 petaFLOP/s in FP64. Grace is a general-purpose processor with high memory bandwidth that delivers excellent floating-point and integer performance across a wide range of applications [5, 15]. It also maintains binary compatibility with the rest of Beta. Each node includes 144 cores of ARM Neoverse V2 at 3.5Hz (8 teraFLOP/s FP64) with 0.9 Tbyte

of LPDDR5X memory (0.95 Tbyte/s). Alpha will also be expanded with 6 HGX H200 nodes (48 H200 GPUs). Note that Alpha employs x86 CPUs. Thus, between Alpha, the Grace-Grace nodes, and the SuperPOD, researchers can pick the tool best suited for their needs.

6 Future facility and technology roadmap

The conceptual design of the final facility is underway with expected completion in Summer 2025. The design goal is 5MW for the first phase of operations, with cost-efficient incremental upgrade to 15MW. The facility's location on the SUNY UB campus was required in the state funding commitment. This geographic location is considered strategic for the sustainability and operational cost efficiency of the facility. It will be powered by Niagara Falls hydroelectric power [11] and will eventually be cooled with water from the UB central chilled water plant, to which warm water will be returned, enabling recovery of heat to warm buildings in the extended cold winter. Very efficient dry coolers will provide resilience and cover summer peak loads.

To track technology, software, and algorithms, we presently plan to install over the next 10 years three generations of computer in the new facility, each at circa $100M. The first such system, Gamma, is expected to start operations by spring 2027 and should be at least 3x more capable than Beta. We have so far made no technology selection for Gamma, with our future experience with Beta being highly relevant.

7 Preliminary economic impact

In March 2025, HR&A Advisors, Inc. performed a preliminary economic impact study using IMPLAN's input-output model and multipliers [10] of Empire AI's initial investments in Alpha, Beta, construction of the future computer facility, and its first generation of computers that together will total about $196 million.

- Together, these investments will generate a one-time total impact of 750 job-years, $67 million in labor income, and $155 million in economic output in New York State between 2024 and 2026.
- The Alpha and Beta clusters will generate 220 job-years, $20 million in labor income, and nearly $46 million in economic output.
- The investment into the permanent Center will generate a total impact of 530 job-years, $47 million in labor income, and $109 million in economic output in New York State between Q2 2025 and Q4 2026.

These are preliminary projections will continue to be refined, and do not account for economic impacts generated through future operations or associated research activity, inventions produced, startups created, faculty and student recruitment, and other activities.

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